Grade 12 Essential Mathematics

Teachers' Book

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INNODEMS

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Preface

Introduction. This is a free interactive digital textbook following the Grade 12 Kenyan competency based maths curriculum and has been co-developed by INNODEMS, IDEMS and the Kenyan Maths Society. It comes with teaching resources such as lesson plans and step-by-step guides to support teachers teaching this curriculum for the first time. You can find these in the respective sections of the book.

Interactive elements of the online book include e.g. automatically evaluated checkpoint questions (see further below) and animations (TODO) that can be projected in class or used by learners at home to explore on their own.

This is the print version. You can find the interactive online version here: Online version.

Support. We also offer a free online course on teaching using competency-based education methodologies and contributing to these learning resources: CBE Math Delivery I Course

Access the course directly using the link above, or follow this guide to install the Moodle App to more easily follow the course on your phone.

We also offer free online trainings and opportunities to exchange you experience with other teachers. The strand-specific courses are linked in the sections of this book. You can find more information in the resources section of the project website.

For Learners. Click here for an (online) learner version of the book. Share this with learners (or their parents) who have access to digital devices. It features interactive **Checkpoint** questions that learners can use to practice at their own pace and get instant feedback on their solutions. These questions are randomized, so that learners can attempt a question multiple times and get a different random variant each time.

Checkpoints: An example. Below you can find an example of a checkpoint question. While in print these look like regular exercises, in the online version these are interactive. Learners can enter their answers in the input fields and get immediate feedback. They can also attempt new random variants of the questions to practice a given concept over and over.

Checkpoint 0.0.1 Exercise

Solve the quadratic equation $x^2 - x - 2 = 0$. If there is a repeated root, enter this twice.

$x =$ _____ or $x =$ _____

Answer 1. 2

Answer 2. -1

Solution. We are asked to factorise the quadratic equation $x^2 - x - 2 = 0$.

First factorise the left-hand side:

$$x^2 - x - 2 = (x - 2)(x + 1).$$

Equate each factor to zero to obtain the two solutions:

Solution 1: $x = 2$

Solution 2: $x = -1$.

Depending on your classroom setup, you also can make use of such checkpoints in your teaching in different ways. If your learners have access to digital devices during class, they can use the online version of the book to practice, but you can also use these to generate different versions of a question for group

work or quizzes without devices, or to project a question on the board and automatically verify learners' answers and give feedback.

Syllabus Alignment. The content of this book is aligned with the Kenyan Grade 12 Competency Based Curriculum for Mathematics. In Appendix A and Appendix B you can find a detailed mapping of the content of this book to the curriculum, including which learning outcomes are covered in which sections of the book.

Sections marked with an asterisk (∗) in the title indicate sections that cover content that is not explicitly mentioned in the syllabus, but that we consider important for learners to master.

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Part I

Numbers and Algebra

Chapter 1

Sets

1.1 Set Notation

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Identify set notations and elements of a set in varied situations

1.2 Representation of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Represent sets in different situations

1.3 Types of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Identify set notations and elements of a set in varied situations

1.4 Operations on Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Perform operations on sets in different situations

1.5 Venn Diagrams

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Draw Venn diagrams to represent sets

1.6 Application of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Apply sets to real life situations

Chapter 2

Compound Proportions, Rates of Work and Mixtures

2.1 Proportional Parts

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Divide quantities using ratios in different situations

2.2 Compound Proportions

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Determine direct, indirect and inverse proportions from real life situations; Determine compound proportions using unitary method

2.3 Rates of Work

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Compute rates of work from real life situations

2.4 Mixtures

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Determine components of mixtures in different situations

Chapter 3

Linear Inequalities

3.1 Simple and Compound Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Linear Inequalities
Specific Learning Outcomes	Form simple inequalities in one unknown from real life scenarios; Solve simple and compound inequalities in one unknown

3.2 Linear Inequalities in Two Unknowns

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Linear Inequalities
Specific Learning Outcomes	Form linear inequalities in two unknowns from real life scenarios

3.3 Graphical Representation of Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Linear Inequalities
Specific Learning Outcomes	Represent linear inequalities in two unknowns graphically

3.4 Graphical Solutions of Simultaneous Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Linear Inequalities
Specific Learning Outcomes	Represent linear inequalities in two unknowns graphically

3.5 Forming Linear Inequalities from Graphs

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Linear Inequalities
Specific Learning Outcomes	Form simple linear inequalities from inequality graphs

Chapter 4

Linear Programming

4.1 Forming Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Form linear inequalities from real life scenarios

4.2 Graphing Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Draw linear inequalities on a cartesian plane

4.3 Optimisation of Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Determine optimal solutions using sampling technique

4.4 Application of Linear Programming

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Apply linear programming to real life situations

Part II

Measurements and Geometry

Chapter 5

Area Approximations

5.1 Counting Squares

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Area Approximations
Specific Learning Outcomes	Use counting squares technique to estimate area of irregular shapes and curves

5.2 Trapezoidal Rule

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Area Approximations
Specific Learning Outcomes	Use trapezoidal rule to estimate area bounded by a curve and lines

5.3 Mid-ordinate Rule

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Area Approximations
Specific Learning Outcomes	Apply mid ordinate rule to estimate area bounded by a curve and lines

Chapter 6

Vectors

6.1 Vector and Scalar Quantities

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Identify vector and scalar quantities in different situations

6.2 Vector Notation

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Represent vectors geometrically on Cartesian plane

6.3 Representation of Vectors

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Represent vectors geometrically on Cartesian plane

6.4 Addition of Vectors

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Add vectors in different situations

6.5 Multiplication of a Vector by a Scalar

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Multiply vectors by scalars

6.6 Magnitude of a Vector

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Determine the magnitude of a vector using cartesian coordinates

6.7 Translation Vector

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Vectors
Specific Learning Outcomes	Draw an image of an object under a translation vector on a Cartesian plane

Chapter 7

Navigation

7.1 Longitudes and Latitudes

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Navigation
Specific Learning Outcomes	Identify longitudes and latitudes in different resources; Locate points on the earth surface using latitudes and longitudes

7.2 Distances on the Earth's Surface

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Navigation
Specific Learning Outcomes	Determine distance along parallels of longitude and latitude in kilometres and in nautical miles

7.3 Distance and Speed

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Navigation
Specific Learning Outcomes	Determine speed in knots and in kilometres per hour

7.4 Longitude and Time

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Navigation
Specific Learning Outcomes	Determine local time on the earth's surface using longitudinal difference

Chapter 8

Commercial Arithmetic III

8.1 Simple Interest

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Compute simple interest in financial transactions

8.2 Compound Interest

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Determine compound interest using the formula

8.3 Appreciation

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Compute appreciation and depreciation in value of assets

8.4 Depreciation

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Compute appreciation and depreciation in value of assets

8.5 Hire Purchase

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Determine hire purchase costs of goods

8.6 Income Tax

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Commercial Arithmetic III
Specific Learning Outcomes	Compute net income for employees using taxation table

Part III

Statistics and Probability

Chapter 9

Statistics III

9.1 Mean, Median and Quartiles by Calculation

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics III
Specific Learning Outcomes	Determine mean, variance and standard deviation for ungrouped data

9.2 Cumulative Frequency Curves

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics III
Specific Learning Outcomes	Draw a cumulative frequency curve for a set of data

9.3 Median and Quartiles for Grouped Data

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics III
Specific Learning Outcomes	Calculate median and quartiles for ungrouped and grouped data; Estimate median, quartiles, interquartile range and quartile deviation from cumulative frequency curve

9.4 Measures of Dispersion

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics III
Specific Learning Outcomes	Determine mean, variance and standard deviation for grouped data; Estimate median, quartiles, interquartile range and quartile deviation from cumulative frequency curve

Chapter 10

Probability III

10.1 Theoretical Probability

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.2 Probability III
Specific Learning Outcomes	Determine discrete probabilities in different situations

10.2 Discrete Probability

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.2 Probability III
Specific Learning Outcomes	Determine discrete probabilities in different situations

10.3 Laws of Probability

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.2 Probability III
Specific Learning Outcomes	Apply laws of probability to predict likelihood of events

10.4 Combined Events

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.2 Probability III
Specific Learning Outcomes	Determine probability of mutually exclusive and independent events

10.5 Tree Diagrams

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.2 Probability III
Specific Learning Outcomes	Calculate probabilities using tree diagrams

Appendices

Appendix A

Syllabus Alignment

This section shows how the content of this book aligns with the Kenyan Grade 12 competency-based mathematics curriculum. Use the links to navigate to the relevant sections.

Appendix B

Learning Outcomes Coverage Table

For each strand and sub-strand in the 2025 Core Mathematics CBE Curriculum, this table maps each specific learning outcome to the section(s) of the textbook where it is covered. Section numbers are clickable links.

Colophon

This book was authored in Pre-
TeXt by the following contributing team: